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ORIGINAL ARTICLE

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Reducing lung cancer risk from radon exposure in Manheim Township, PA: Radon Resistant New Construction (RRNC) as a significant public health tool

Yiguang Zhu[a], Mary A. Fox[b], Bryce Corrigan[c] and Paul A. Locke[a]

[a]Department of Environmental Health and Engineering, Johns Hopkins Bloomberg School of Public Health, Baltimore, MD, USA; [b]Department of Health Policy and Management, Risk Sciences and Public Policy Institute, Johns Hopkins Bloomberg School of Public Health, Baltimore, MD, USA; [c]SNF Agora Institute, Johns Hopkins University, Baltimore, MD, USA

ABSTRACT

Purpose: Radon is a known human carcinogen and the second leading cause of lung cancer worldwide. This study evaluated the effectiveness of Radon Resistant New Construction (RRNC) ordinances enacted in Manheim Township, Pennsylvania, which mandate RRNC features in all new dwellings, to reduce lung cancer risk associated with radon exposure.

Materials and Methods: We identified new dwellings in Manheim Township built under RRNC ordinances and analyzed corresponding radon test and mitigation data from PA DEP databases. After carefully reviewing the data and correcting database reporting anomalies, we evaluated indoor radon levels with and without RRNC and estimated the average reduction. Finally, we employed an existing dose-response model to estimate the average reduction of lung cancer risk associated with RRNC, taking smoking status into consideration.

Results: We analyzed 1,734 radon tests conducted in 483 homes, and found on average those homes built with RRNC features reduced the indoor radon concentration by 16.71 pCi/L and lowered the excess relative lung cancer risk attributable to radon exposure by 99%, regardless of smoking status.

Conclusions: Using real world data collected both before and after a change in municipal laws, these results provide strong evidence supporting the adoption of similar mandatory RRNC in other PA jurisdictions, nationwide, and in other countries that seek to reduce radon exposure and associated risks in indoor environment. This study demonstrates that the adoption and implementation of RRNC can reduce indoor radon levels, and consequently lower radon-related lung cancer risks.

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Indoor radon; radon risk analysis; lung cancer

Introduction

Radon is a colorless, odorless, radioactive gas that is generated from the natural decay of uranium and radium, which are ubiquitously present in almost all rocks and soils (NRC 1999). Radon typically enters homes and other buildings through cracks and openings in foundations, floors, or walls in contact with the ground, subsequently accumulating in enclosed spaces. Other less common pathways include off-gassing from some building materials or radon dissolved in groundwater (EPA 2012). Indoor radon will spontaneously decay into electrically charged radon progenies, attach to particulate matter in the air, and be inhaled. Once inhaled, the radon gas itself is largely exhaled, but the short-lived radon progenies can deposit in the lung epithelium. These deposited progenies will continuously decay and emit alpha radiation, which can irradiate sensitive cells and potentially cause lung cancer (NRC 1999; Tirmarche 2018). Its significance as a public health concern is recognized globally, with major guidance provided by multiple international organizations (WHO 2009; ICRP 2014).

The association between cumulative radon exposure and increasing lung cancer risk in the general population was determined based on data from cohort studies of uranium miners (NRC 1999). Pooled case-control studies around the world confirmed this analysis, and found residential radon exposure at lower doses yielded consistent dose response information (Lubin, Wang, et al. 2004; Darby et al. 2005; Krewski et al. 2006). A global estimation conducted in 2012 indicated that radon exposure was implicated in more than 220,000 deaths in 66 countries (Gaskin et al. 2018). In the United States, the Environmental Protection Agency (EPA) indicates that radon is associated with about 21,000 lung cancer deaths every year, approximately 18,000 in smokers, and 3,000 in nonsmokers (EPA 2022a). Current smokers and ex-smokers are at higher risk when compared to nonsmokers, because the biological mechanisms leading to the development of lung cancer will be sub-multiplicatively increased when humans are impacted by both radon exposure and smoking (NRC 1999).

Radon-resistant new construction (RRNC) is an indoor radon mitigation technique that has been proven to be

CONTACT Yiguang Zhu yzhu99@jh.edu Department of Environmental Health and Engineering, Johns Hopkins Bloomberg School of Public Health, 3400 N. Charles Street, Baltimore, MD 21218, USA.
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effective (ANSI/AARST 2020) in reducing radon levels in buildings. RRNC can be categorized into active and passive sub-slab depressurization systems, both use gas-permeable layers and sealants to insulate radon emitted from underlying soils and rocks and use air-handling units to vent radon gas from the structure. Passive systems connect the sub-slab area to outdoor air via vent pipes and draw radon beneath the slab by pressure difference, while active systems are equipped with fan-powered vents (ICC 2018). The radon removal effectiveness of both passive and active RRNC has been extensively demonstrated in the literature (LaFollette and Dickey 2001; Rahman and Tracy 2009). Although RRNC can reduce the radon level to below 2 pCi/L, it has not been shown to decrease the radon level to the outdoor level (EPA 1994).

In the United States laws and regulations on indoor radon have been enacted at the federal level. The Indoor Radon Abatement Act (IRAA), passed in 1988, authorized EPA to establish grant programs, promulgate some regulations and publish a citizen's guide regarding the indoor radon (15 U.S.C. §2661 1988). The Indoor Air Quality and Radon Gas Act of 1986 was codified in a note to the U.S. Code and guided EPA to establish a research program about indoor air quality, which included the study of radon (Bellamy and Locke 2018). However, no federal radon regulation has been promulgated by EPA using the given authorities. Promotion programs such as Federal Radon Action Plan led by EPA have not achieved sufficient progress in controlling indoor radon and were gradually phased out (Gordon et al. 2018).

In the absence of federal regulation, the Commonwealth of Pennsylvania (PA), which was ranked as the state with the highest average indoor radon level in the United States, has taken action to protect its citizens against indoor radon hazards by enacting a series of laws (Lewis and Houle 2009). For example, the Radon Certification Act required the PA Department of Environmental Protection (DEP) to certify radon testers and mitigators and collect radon test records. Additionally, 13 jurisdictions in PA passed ordinances that required RRNC in all new dwellings. Manheim Township (Twp) located in Lancaster County, has the largest population (n = 43,977) and the most households (n = 16,173) among these 13 jurisdictions (U.S. Census Bureau 2022). On June 14th, 2004, Manheim Twp enacted two ordinances that adopted the Appendix F of the International Residential Code and required all the new dwellings to install passive RRNC (Commissioners of Manheim Township 2004a, 2004b; IRC 2018 – APPENDIX F). These two Ordinances will be referred to as the "RRNC Ordinances" in this paper.

While prior studies of PA have established the general effectiveness of RRNC Ordinances by investigating several PA jurisdictions, significant gaps remain in quantifying the real-world impact of such policies on potential health outcomes. Existing studies have often focused primarily on radon level reductions without quantitatively estimating the associated lung cancer risk implications, or have discussed risk only qualitatively. Furthermore, many previous evaluations lacked detailed consideration of smoking status interaction or were based on limited sample sizes (Bellamy and Locke 2018; Edelson-Mammel and Locke 2019; Cronin et al. 2020).

As a result, there is a paucity of studies that use actual measurements in homes with RRNC to assess the impact of local radon control ordinances on public health. This study seeks to partially fill that gap by investigating the impact of the enactment of RRNC ordinances on radon levels and associated lung cancer risk reduction based on the changes in such levels after RRNC is installed. Specifically, we use real-world data of radon measurements in homes to: (1) Quantify the radon reduction achieved by the RRNC ordinance, (2) Estimate the corresponding potential lung cancer risk reduction using the Darby model, in an effort to move beyond qualitative discussion, and (3) Evaluate these potential risk reductions across smoking status using local prevalence data on smoking, acknowledging the critical radon-smoking relationship for lung cancer risk. Therefore, this study provides a data-driven evaluation of the estimated public health impact of a specific, implemented radon control ordinance, highlighting its value as a significant public health tool.

Methods

Study design

We obtained the Manheim Twp new housing information from 2004 to 2018 from the Manheim Twp Commissioners and the PA DEP. After reviewing this data, we found that after the enactment of the RRNC Ordinances on June 14th, 2004, at least 1,733 new dwellings with RRNC features were built or reconstructed in Manheim Twp, and all homes except for 6 were located within two zip codes (17601 & 17543). By eliminating the 6 homes outside these zip codes, we created a "Housing Dataset" that has 1,727 addresses.[1]

Johns Hopkins University and the PA DEP entered into a Collaborative Agreement that provided access to a database with approximately 2,000,000 radon test records (the "Testing Dataset"), and a database with about 192,000 radon mitigation records (the "Mitigation Dataset"). These databases contained radon test measurements, that were collected between 1988 to 2017. PA laws require that certified testers and mitigators licensed to do business in PA report this data to the Commonwealth. These records were submitted to PA DEP via the PA DEP GreenPort web portal. Radon test records had information including indoor radon levels, house styles, test methods, dates, location, etc.; radon mitigation records had information including indoor radon levels, test dates, mitigation methods, costs, etc. It should be noted that many addresses had multiple records, because multiple radon tests were taken at the same location. We filtered these databases according to zip code and by test time to find those homes within the two Manheim Twp zip codes that we had identified by examining the Housing Dataset. Applying these criteria (time of passage of ordinance and zip code) we retained 22,026 tests conducted in 9,002 homes from the Testing Dataset, and 6,628 mitigation records collected from 2,867 addresses from the Mitigation Dataset, for further analysis.

Based on the addresses of the new dwellings in the Housing Dataset, we then carried out an address matching for the Testing Dataset and the Mitigation Dataset. Four

hundred and fifty-nine addresses with 1,234 test records in the Testing Dataset and 213 addresses with 503 mitigation records in the Mitigation Dataset were identified. The records extracted from the Mitigation Dataset were labeled as pre-mitigation or post-mitigation to indicate whether a radon test was taken before or after the installation of an RRNC system. We categorized the records extracted from the Testing Dataset into pre- or post-mitigation based on factors. First, the home had multiple records collected from the same address at different dates. Second, the radon levels had a difference larger than 20%, records with higher radon levels (recorded first) were labeled as pre-mitigation while records with lower radon levels (taken after the higher levels) were labeled as post-mitigation. Records that did not fall into the above criteria were examined on a case-by-case basis. As a result, all 1,234 records adopted from the Testing Dataset were examined and categorized. By combining the matched records from Testing and Mitigation Dataset and eliminating duplications, we created a new dataset that will be referred to as the "Raw RRNC Dataset" in this paper. The Raw RRNC Dataset consisted of 1,734 records collected from 483 Manheim Twp homes, which contained 821 pre-mitigation tests and 913 post-mitigation tests. All these measurements were taken after the enactment and effective date of RRNC Ordinances. In addition, these measurements showed substantial reductions in radon levels, because the first measurement was more than 20% higher than the second measurement. We therefore classified these 483 houses as dwellings that had installed RRNC systems.

However, some houses either had pre- or post-mitigation records (n = 201) – but not both. We excluded these homes from further analysis to focus on those records that had a before and after measurement for the same address. After removing these homes, we found that 1,393 tests were conducted at 285 addresses. For each address, if there were multiple pre- or post-mitigation records in an address, we calculated the arithmetic mean (AM) of the pre- and post-radon levels and used this arithmetic mean for further analysis. As a result, we compiled a dataset with 285 pre-mitigation and 285 post-mitigation radon arithmetic means, which gave us a pre- and post-mitigation measurement for each of the 285 addresses. This database will be referred to as "Final RRNC Dataset". Table 1 displays the summary statistics for the four datasets. Figure 1 illustrates the inclusion and exclusion of the radon test and mitigation records in the four datasets.

To obtain smoking prevalence for the study population, we accessed the Behavioral Risk Factor Surveillance System (BRFSS) embedded in the Enterprise Data Dissemination Informatics Exchange (EDDIE) Platform, a data dissemination tool provided

Table 1. Radon concentrations (pCi/L) and reduction (%), stratified by datasets.

Datasets	n (addresses/records)	Censored records	TAM radon levels[a] (95% CI)	Range	Reduction[b] (%)
Testing Dataset	9,002/22,026	2,632	8.88 (8.59, 9.17)	0.50-650.40	–
Mitigation Dataset	2,867/6,628	1,198	8.08 (7.51, 8.65)	0.50-335.00	21.57 (93.79)
Pre-mitigation	2,681/2,921	6	23.00 (23.92, 22.08)	0.50-335.00	–
Post-mitigation	2,646/3,707	1,192	1.43 (1.49, 1.37)	0.50-49.70	–
Raw RRNC Dataset	483/1,734	274	7.53 (6.79, 8.27)	0.50-118.00	17.23 (94.67)
Pre-mitigation	389/803	0	18.20 (17.14, 19.27)	0.70-118.00	–
Post-mitigation	379/931	274	0.97 (0.88, 1.06)	0.50-9.20	–
Final RRNC Dataset	285/570	–	9.49 (7.99, 11.00)	0.50-100.00	16.71 (93.61)
Pre-mitigation	285/285	–	17.85 (16.21, 19.48)	3.10-100.00	–
Post-mitigation	285/285	–	1.14 (1.03, 1.25)	0.50-8.75	–

Note: TAM, Tobit Adjusted Mean; RRNC, radon-resistant new construction.
[a]Tobit mean is Tobit regression adjusted arithmetic mean. For Final RRNC Dataset, the arithmetic means of radon levels were reported.
[b]Estimate by subtracting post-mitigation mean radon levels from pre-mitigation mean radon levels. There is no result reported for Testing Dataset because pre- or post-mitigation information is not available.

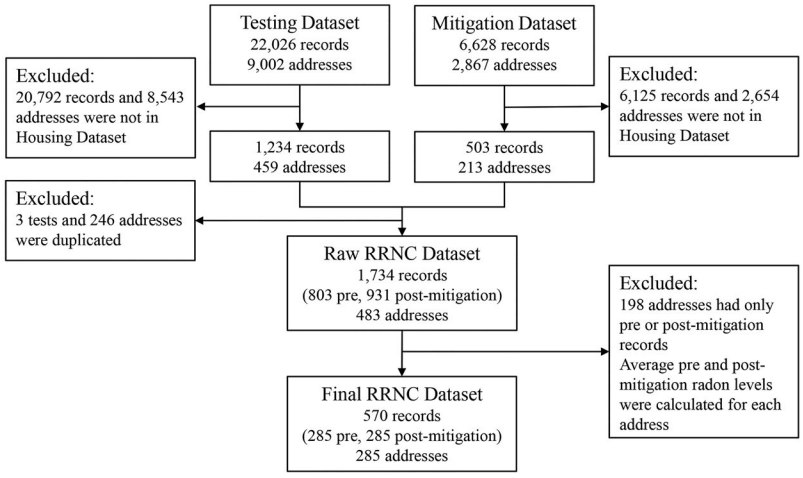


Figure 1. Flow diagram of radon test and mitigation records included in the test, mitigation, raw RRNC, and Final RRNC datasets. RRNC, radon-resistant new construction.

by the PA Department of Health. The best available behavioral risk factor data were country-level data. Since Manheim Twp is within Lancaster County, we obtained the smoking prevalence data for Lancaster County. The EDDIE Platform provided the smoking prevalence on a 3-year basis. From 2016 to 2018, among all the adults in Lancaster County, 11% [95% Confidence Interval (CI): 7%, 15%] were current smokers, 24% [95% CI: 19%, 29%] were ex-smokers, and 66% [95% CI: 60%, 71%] were nonsmokers (EDDIE Platform 2022).

Statistical methodologies

As mentioned earlier, all certified testers and mitigators are required by PA law to report testing data to the GreenPort web portal. This web portal truncates low radon measurements so that all tests taken that recorded levels at or below 0.5 pCi/L are entered as 0.5 pCi/L. As a result, all test results lower than 0.5 pCi/L were rounded up to 0.5 pCi/L. This artificially imposed lower reporting limit means that reported data is left-censored (Helsel 1990). As a result, a substantial number of data points at 0.5 pCi/L created heavily left-censored distribution. Figure 4 shows this left censoring as a series of histograms. Without correction, this accumulation of data points at 0.5pCi/L would impact the accuracy of the data analysis and as a result we sought to address this issue by applying Tobit regression. Tobit regression is a well-accepted approach to treating left-censored data in many disciplines, including in environmental, public health, and epidemiological studies (Lubin, Colt, et al. 2004; Smith et al. 2006). It can be used to estimate summary statistics by utilizing the reporting limit and the underlying distribution (Helsel 1990). When the reporting limit is known and the proportion of data below the reporting limit is less than 30%, Tobit regression can provide a relatively more accurate estimation when compared to other common methods (Uh et al. 2008). After preliminary assessment, censored observations in our datasets ranged from about 12% to 30%, thus Tobit regression was considered suitable for the analysis of our datasets. The numbers of censored records for each dataset were reported in Table 1.

To establish a general background of the indoor radon level in Manheim Twp, we first analyzed the distribution of records in the Testing Dataset and computed Tobit regression adjusted arithmetic mean (TAM) radon levels. We also assessed the distribution of natural log-transformed radon levels to check normality. To adjust for the impact of the censored data, we also explored if the data followed a truncated normal distribution. The distribution of indoor radon levels and log-transformed radon levels are shown in Figure 4. The Raw RRNC Dataset was analyzed using the same procedures. We compared the outcomes of the Testing and Raw RRNC Datasets to see if the radon levels in all dwellings and new dwellings have significant differences. We then computed the TAM radon levels for pre- and post-mitigation records in the Mitigation and Raw RRNC Datasets. By subtracting post-mitigation TAM radon levels from pre-mitigation TAM radon levels, we estimated the reduction of indoor radon levels associated with RRNC. In the analysis of the Final RRNC Database, we used AM rather than TAM radon levels for pre-mitigation records because these were substantially above the reporting limit of 0.5 pCi/L. To maintain methodological consistency, we also applied AM to the post-mitigation records. This approach ensures uniformity in our analysis, despite the presence of censored data points in the post-mitigation records. Other than that, the same method was used for the analysis of Final RRNC Database. The reductions of indoor radon level were reported in Table 1.

STATA 16, a statistical software commonly used in public health research, was used to analyze the databases. Tobit regression is a built-in function that can report the number of censored observations, TAM values, and 95% confidence intervals. STATA 16 was also used to perform natural logarithmic transformation.

Dose-response model

By using the dose-response model developed by Darby et al., we evaluated the lung cancer risk reduction due to the reduction in radon levels associated with RRNC installation in homes. The Darby studies were developed based on a combined analysis of 13 case-control studies that involved over 21,000 observations from 13 European countries. The Darby models present linear dose-response relationships between lung cancer risk (excess relative risk, absolute risk, and relative risk) and indoor radon concentration respectively for smokers, ex-smokers, and nonsmokers (Darby et al. 2005; 2006). Several other widely used models were also reviewed, including the BEIR VI model (NRC 1999) and the Krewski model (Krewski et al. 2005). The Darby model was selected because it was developed based on a larger sample size and had a sophisticated method for addressing the relationship between smoking and radon exposure. Our analysis applies these established linear relationships directly to the observed average radon reduction; this study does not extrapolate the model beyond its reported concentration range or alter its function. The Darby model consists of three parts:

Model 1: excess relative risk of lung cancer

For the general population, the Darby model provided a simple linear relationship between lung cancer risk and radon exposure: lung cancer risk will increase by 16% per every 2.70 pCi/L (converted from 100 Bq/m^3) increase in radon concentration, regardless of other lung cancer risk factors. To be more specific, for lifelong nonsmokers, this relationship is applicable to both their relative risk and absolute risk, while for smokers and ex-smokers, this relationship can only be used for relative risk estimation associated with radon reduction.

Several factors can explain this relationship. First, the absolute risk of getting lung cancer is a function of lung cancer risk from radon exposure, smoking, and other factors (sex, age, etc.). For nonsmokers, according to the Darby model, the risk from other factors is trivial compared to radon exposure, so, their absolute lung cancer risk is about the same as the relative risk associated with radon exposure, which is both

estimated to increase 16% per every 2.70 pCi/L rise in radon exposure. However, for smokers and former smokers, the main risk factor for lung cancer is smoking, and their relative lung cancer risk from smoking is significantly higher than from radon exposure alone. For smokers, the Darby model estimates the excess relative risk associated with radon exposure, and the relationship is about a 16% increase in relative risk per every 2.70 pCi/L increase in radon concentration.

Model 2: the cumulative absolute risk of death from lung cancer

As summarized in Table 2, key point estimations of cumulative absolute risk of death from lung cancer by age 75 for current smokers and lifelong nonsmokers at radon levels of 0 pCi/L, 2.70 pCi/L, 5.41 pCi/L (converted from 200 Bq/m^3), 10.81 pCi/L (converted from 400 Bq/m3), and 21.62 pCi/L (converted from 800 Bq/m3) were extracted from the supplement publication by Darby (Darby et al. 2006). In Figure 2, these extracted data points were plotted to visualize the dose-response relationship, and best-fitting straight lines were estimated by ordinary least-squares regression (Microsoft Excel) for current smokers and lifelong smokers. The linear equations that described the relation between cumulative absolute risk of death from lung cancer and radon exposure are respectively listed as Equations (1) and (2):

$$\text{Current Smokers : Cumulative absolute risk}(\%) = 0.0053 \times \text{Indoor radon level} + 0.10 \quad (1)$$

$$\text{Lifelong nonsmokers : Cumulative absolute risk}(\%) = 0.00024 \times \text{Indoor radon level} + 0.0041 \quad (2)$$

Here, the cumulative absolute risk of death from lung cancer reflects the deaths caused by lung cancer by the age of 75. For example, if the cumulative absolute risk of death from lung cancer by age 75 for current smokers is 10% at 5 pCi/L radon exposure, the plain explanation is that, if exposed to 5 pCi/L of radon, 10 in 100 current smokers will die by age 75 from lung cancer (Sainani 2012). Theoretically, given the population, smoking prevalence, and absolute risk of death from lung cancer, the number of deaths in a certain smoking group due to lung cancer can be roughly calculated by Equation (3):

$$\text{Population} \times \text{Smoking prevalence} \times \text{Absolute risk of death} = \text{Lung cancer death toll} \quad (3)$$

Table 2. Point estimations for cumulative absolute risk (%) of death from lung cancer by age of 75 for current smokers and lifelong nonsmokers, according to indoor radon levels.

	Cumulative absolute risk (%) (95% CI)	
Radon Level (pCi/L)	Current smokers	Lifelong nonsmokers
0	10.11	0.41
2.70	11.63 (10.59, 13.03)	0.47 (0.43, 0.54)
5.41	13.12 (11.06, 15.85)	0.54 (0.45, 0.66)
10.81	16.03 (15.11, 21.44)	0.67 (0.49, 0.91)
21.62	21.57 (13.86, 30.98)	0.93 (0.57, 1.42)

Note: Point estimations and 95% CI are extracted from Darby et al. (2006). Current smokers are defined as continuing smokers of 15–24 cigarettes per day. CI, confidence interval.

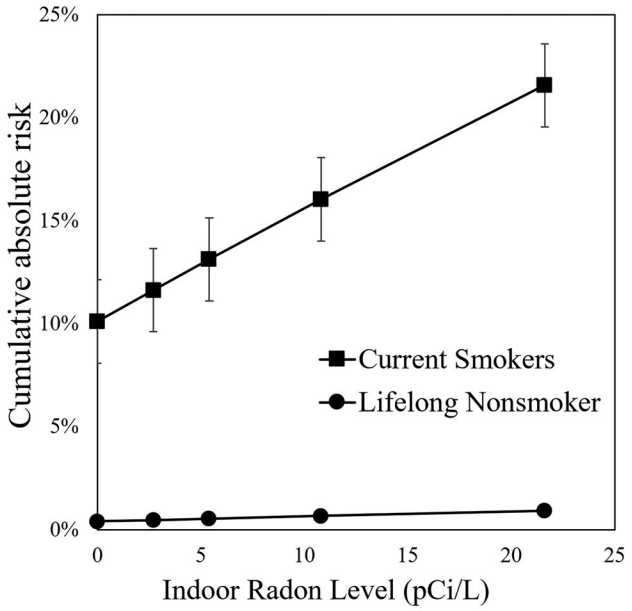


Figure 2. Cumulative absolute risk of death from lung cancer by age of 75 according to indoor radon level for current smokers and lifelong nonsmokers, with best fitting straight lines. Data points extracted from Darby et al. (2006). Current smokers are defined as continuing smokers of 15–24 cigarettes per day.

Model 3: relative risk of lung cancer

As summarized in Table 3, key point estimations of relative lung cancer risk associated with radon exposure for current and ex-smokers who quit more than 10 years ago and within 10 years, at radon levels of 0 pCi/L, 2.70 pCi/L, 5.41 pCi/L, 10.81 pCi/L, and 21.62 pCi/L were extracted from the supplement publication by Darby (Darby et al. 2006). The smoking data on the EDDIE Platform had no information on when ex-smokers quit smoking, so, we calculated average values for the two ex-smoker groups at each radon level to represent the relative risk of the entire ex-smoker group. As shown in Figure 3, these extracted data points were plotted by the authors to visualize the dose-response relationship, and best-fitting straight lines were estimated by ordinary least-squares regression (Microsoft Excel). The dashed reference line at relative risk equal to 1 indicated that lung cancer risks here were calculated relative to lifelong nonsmokers with no radon exposure. The linear equations that described the relation between relative lung cancer risks and radon exposure are displayed as Equations (4) and (5):

$$\text{Current smokers : Relative risk} = 1.53 \times \text{Indoor radon level} + 25.78 \quad (4)$$

$$\text{Ex} - \text{smokers : Relative risk} = 0.77 \times \text{Indoor radon level} + 12.92 \quad (5)$$

Table 3. Point estimations of relative lung cancer risk for current smokers and ex-smokers, according to indoor radon levels.

Relative risk[a] (95% CI)				
Radon Level (pCi/L)	Current smokers	Ex-smokers[b]	Ex-smokers (<10 yrs)	Ex-smokers (≥10 yrs)
0	25.8	12.90	20.8	5.0
2.70	29.9 (27.1, 33.8)	15.00	24.2 (21.9, 27.3)	5.8 (5.2, 6.5)
5.41	34.0 (28.3, 41.7)	17.05	27.5 (22.9, 33.8)	6.6 (5.5, 8.1)
10.81	42.3 (30.9, 49.3)	21.20	34.2 (25.0, 46.7)	8.2 (6.0, 11.2)
21.62	58.8 (36.1, 89.7)	29.45	47.5 (29.2, 72.5)	11.4 (7.0, 17.4)

Note: Point estimations and 95% CI were extracted from Darby et al. (2006). Current smokers were defined as continuing smokers of 15–24 cigarettes per day. CI, confidence interval; yrs, years.

[a]The risks of lung cancer were calculated relative to lifelong nonsmokers with no radon exposure.

[b]Relative risks for ex-smokers were estimated based on the average excess relative risks for ex-smoker who quitted more than 10 years ago and within 10 years.

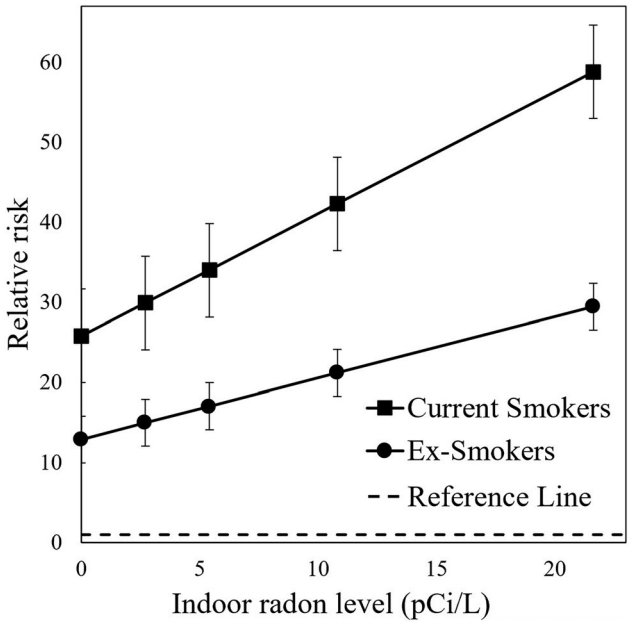


Figure 3. Relative lung cancer risk for current smokers and ex-smokers according to radon exposure, with best fitting straight lines. Reference line indicates that the absolute lung cancer risk for lifelong nonsmokers with no radon exposure is 1. Current smokers are defined as continuing smokers of 15–24 cigarettes per day. Data points extracted from Darby et al. (2006). Data points for ex-smokers were estimated based on the average excess relative risks for ex-smoker who quitted more than 10 years ago and within 10 years.

Relative risk is the ratio of the probability of an outcome in an exposed group to the probability of an outcome in an unexposed group. The unexposed group here is lifelong nonsmokers with no radon exposure, and the exposure groups are the smoker group and ex-smoker group with no radon exposure. The absolute lung cancer risk for the unexposed group is assumed to be 1. Theoretically, the relative lung cancer risks for the exposed groups can be described as multiples of the absolute risk of the unexposed group. For example, if the relative risk of lung cancer for current smokers is 10 at 5 pCi/L radon exposure, the plain explanation is that, if exposed to 5 pCi/L of radon, current smokers are 10 times more likely to get lung cancer than lifelong nonsmokers with no radon exposure (Sainani 2012).

Results

Reduction of indoor radon levels in Manheim Twp

TAM radon levels for the Test and Raw RRNC Datasets are displayed in Table 1. The TAM radon level of Testing Dataset (n = 22,026 records and n = 9,002 addresses) was 8.88 (95% CI: 8.59, 9.17). The TAM radon level of Raw RRNC Dataset (n = 1,734 records and n = 483 addresses) was 7.53 (95% CI: 7.51, 8.65). No significant differences were found when the Testing and Raw RRNC Datasets were compared based on TAM radon levels. Figure 4 illustrates the distribution of indoor radon levels natural logarithm transformed indoor radon levels in Testing and Raw RRNC Datasets. As discussed earlier, the collection of radon measurements that appeared in the lower ranges of all four distribution figures were caused by an artificially imposed reporting limit at 0.5 pCi/L and led to the left-censored distributions in the Testing and Raw RRNC Databases. Tobit regression was applied to treat the censored data and computed corrected TAM values. In both datasets, some test results reported extremely high radon levels, so we set an upper limit at 50 pCi/L when graphing Figure 4A and C. This upper limit did not impact the data analysis. Gaussian curves were plotted in Figure 4B and D to assess normality, it was clear that both distributions were heavily left-censored and were not normally distributed. However, if we set a lower limit at natural log-transformed radon level equal to 1, the log transformed data does follow a truncated normal distribution.

TAM radon levels for pre-mitigation and post-mitigation records in the Mitigation (n = 6,628 records and n = 2,867 addresses), Raw RRNC, and AM radon levels for the Final RRNC Datasets (n = 570 records and n = 285 addresses) are shown in Table 1. By subtracting post-mitigation radon levels from pre-mitigation radon levels, the reduction of indoor radon levels associated with RRNC was estimated for each of the three datasets. The reduction in the Mitigation dataset was 21.57 pCi/L, in the Raw RRNC Dataset was 17.23 pCi/L, and in the Final RRNC Dataset was 16.71 pCi/L. In the Mitigation and Raw RRNC Dataset, because some addresses only had pre- or post-mitigation records, when we calculated the radon reduction by subtracting the two values, the missing values were substituted by zero, which would undermine the statistical significance of the analysis. For the Final RRNC Dataset, each address had corresponding pre- and post-mitigation records, and the subtraction was done on a house-by-house basis, making this estimation most representative of real-world conditions. Hence, the reduction of 16.71 pCi/L in this dataset is considered most indicative of the effectiveness of RRNC. The consistency of reductions across these datasets reinforces our confidence in the reliability of the chosen methodology.

Reduction of lung cancer risk in Manheim Twp

Figure 5 illustrates the reduction of excess relative risk of lung cancer associated with RRNC. According to model 1, the excess relative risk associated with radon exposure would decrease by 16% per every 2.70 pCi/L decrease in

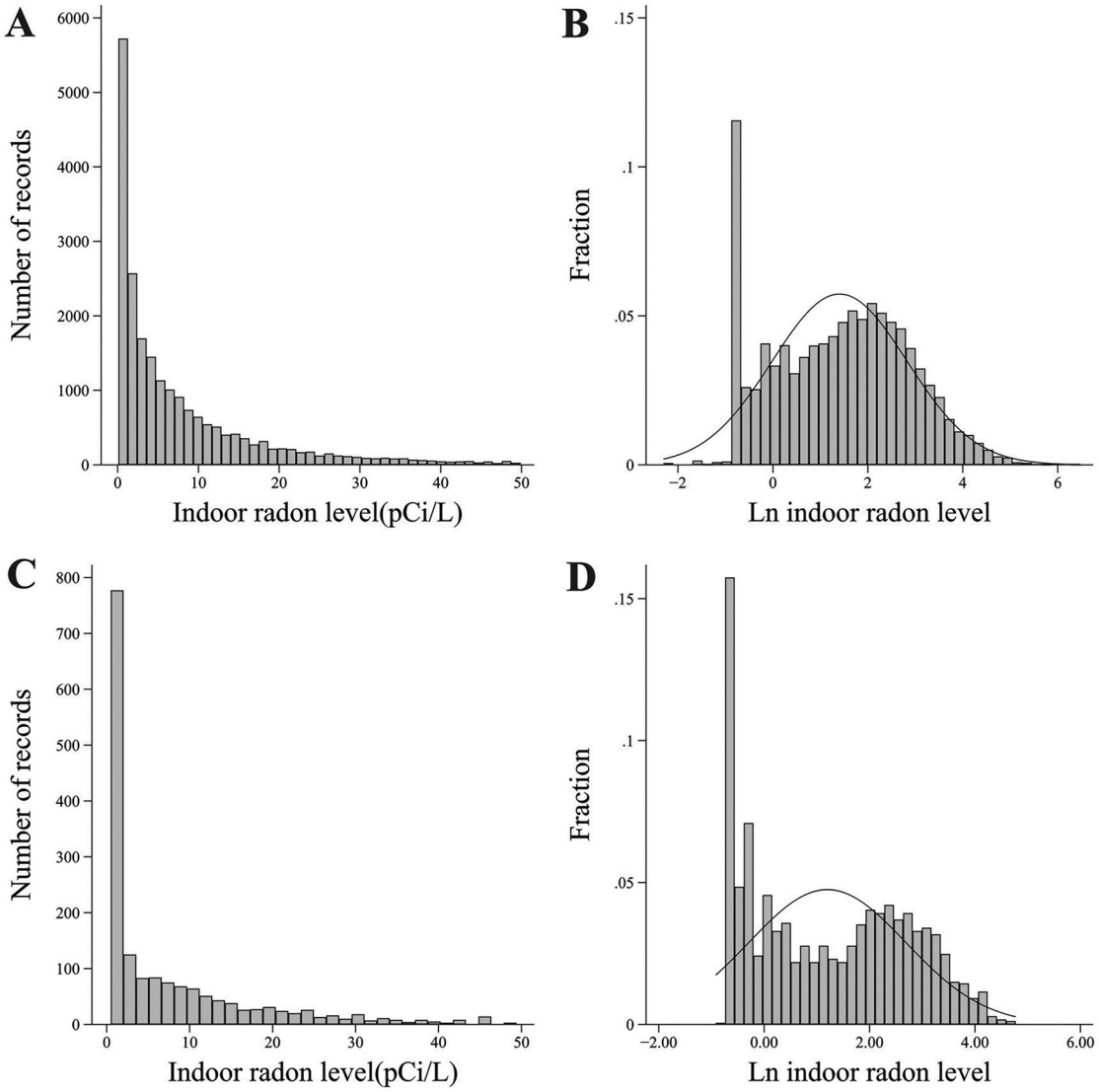


Figure 4. (A) Distribution of indoor radon levels (< 50 pCi/L) of records in Testing Dataset (n = 22,026). (B) Distribution of natural logarithms transformed indoor radon levels of records in Testing Dataset. (C) Distribution of indoor radon levels (< 50 pCi/L) of records in Raw RRNC Dataset (n = 1,734). (D) Distribution of natural logarithms transformed indoor radon levels of records in Raw RRNC Dataset. Ln, natural logarithms transformation.

indoor radon level. From the Reduction Database, we estimated that, on average, RRNC lowers radon levels by 16.71 pCi/L in Manheim Twp homes. By substituting in the radon reduction value into the linear relationship, we found that in Manheim Twp, RRNC was associated with an 99.0% reduction in the excess relative risk of lung cancer for the general population, regardless of smoking status. In other words, RRNC installation largely eliminated the excess relative lung cancer risk associated with radon exposure for nonsmokers. It is important to clarify that this 99% reduction in the overall excess lung cancer risk reflects the large average decrease in radon concentration achieved, and that the lung cancer risk of active and ex-smokers is going to be higher because of the role that smoking plays in raising the risk of lung cancer.

As shown in Figure 6, by using Model 2 (equations 1 and 2) to assess the health impact of the radon reduction, the cumulative absolute risk of death by the age of 75 for current smokers (n = 11% of all adults) decreased from 19.64% to 10.81%, which was a 39.27% reduction; for lifelong nonsmokers (n = 66% of all adults), decreased from 0.84% to 0.44%, a 41.88% reduction. Given the absolute risk reductions, if we assume all dwellings in Manheim Twp have RRNC, we could use Equation 3 to roughly estimate the avoided lung cancer death cases:

$$\text{For current smokers}: 43{,}997 \times 11\% \times (19.64\% - 10.81\%) \approx 427$$

$$\text{For nonsmokers}: 43{,}997 \times 66\% \times (0.84\% - 0.44\%) \approx 116$$

As of 2020, the population of Manheim Twp is 43,997. According to data collected from 2016 to 2018, 11% and 66% are the prevalence of current smokers and life-long nonsmokers among all adults in Manheim Twp (EDDIE Platform 2022; U.S. Census Bureau 2022). Theoretically, the enactment and implementation of the Manheim Twp RRNC Ordinances potentially avoided 427 lung cancer deaths among smokers and 116 lung cancer deaths among lifelong nonsmokers.

The results obtained by Equations 4 and 5 can be seen in Figure 7. Compared to lifelong nonsmokers with no radon

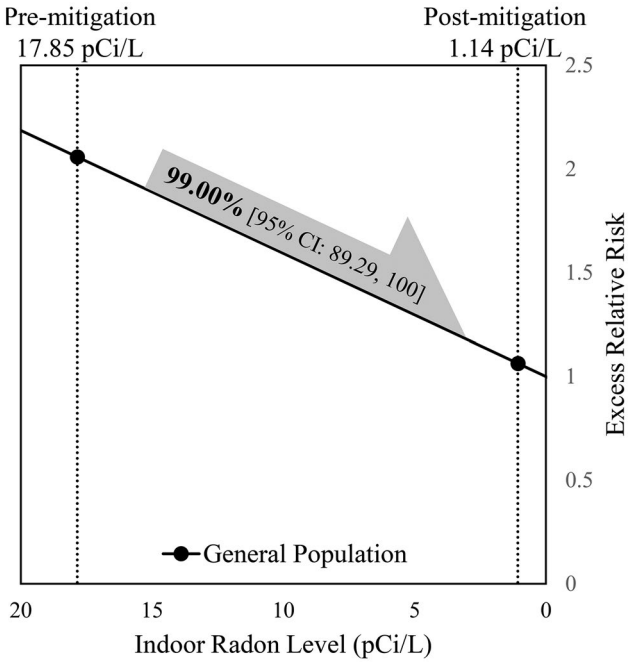


Figure 5. Reduction of excess relative risk of lung cancer associated with RRNC for the general population in Manheim Township, regardless of smoking status. Two vertical dashed lines indicate the TAM radon levels pre and post mitigation. The x-axis is in reverse order.

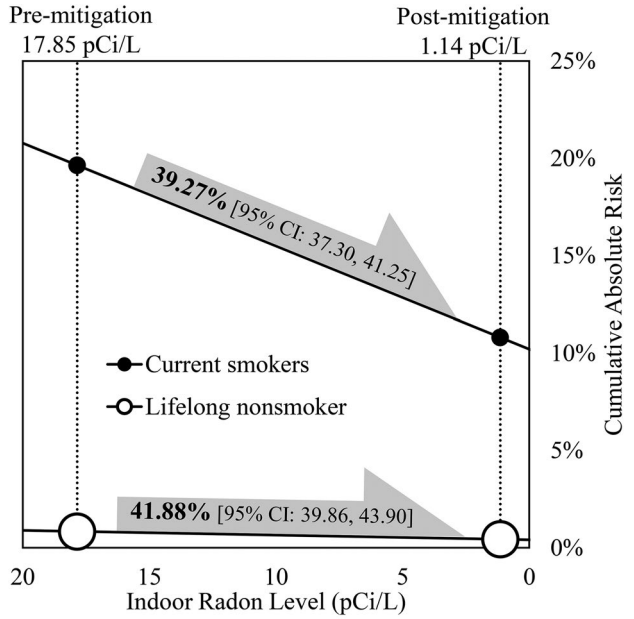


Figure 6. Reduction of cumulative absolute risk of death from lung cancer associated with RRNC by the age of 75 for current smokers (n = 11% of all adults) and lifelong nonsmokers (n = 66% of all adults) in Manheim Township. Areas of the circles represent the proportion of smoking status. Two vertical dashed lines indicate the TAM radon levels pre and post mitigation. The x-axis is in reverse order.

exposure, without RRNC, current smokers were 53.04 times and ex-smokers were 27.52 times more likely to get lung cancer; with RRNC, the relative risks should be reduced to 26.57 for current smokers and 13.79 for ex-smokers. For both current and ex-smokers, the relative lung cancer risks were reduced by over 48%.

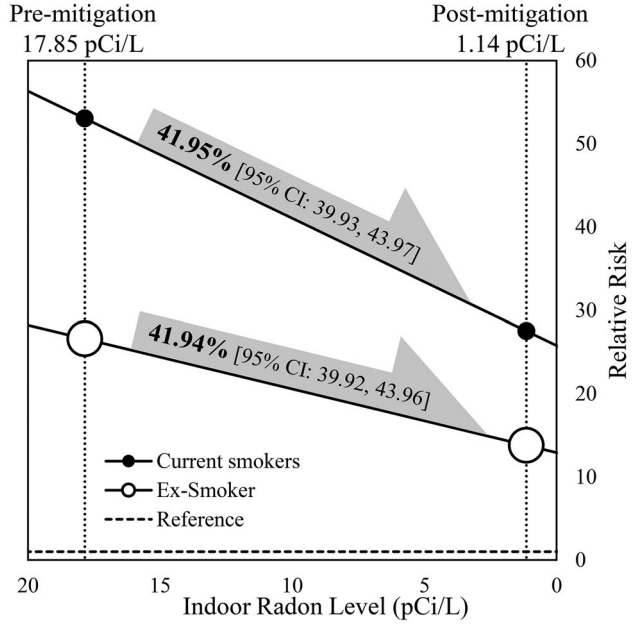


Figure 7. Reduction of relative risk of lung cancer associated with RRNC for current smokers (n = 11% of all adults) and ex-smokers (n = 24% of all adults). Reference line indicates that the absolute lung cancer risk for lifelong nonsmokers with no radon exposure is 1. Two vertical dashed lines indicate the TAM radon levels pre and post mitigation. Areas of the circles represent the proportion of smoking status. The x-axis is in reverse order.

Discussion

In this study, we evaluate the effectiveness of RRNC ordinances in reducing radon levels and estimate the associated reduction in lung cancer risk using the Darby model. These ordinances require RRNC installation in all new dwellings built in Manheim Twp. Radon testing and mitigation records provided by PA DEP were used to estimate the indoor radon levels with and without RRNC and a database provided by Manheim Twp was consulted to obtain addresses of dwellings built with RRNC features. The Darby model was applied to this data to calculate the excess relative risk, cumulative absolute risk, and relative risk of lung cancer for lifelong nonsmokers, ex-smokers, and current smokers (Darby et al. 2005; 2006). Several studies have shown the effectiveness of similar ordinances in terms of radon reduction (Bellamy and Locke 2018; Edelson-Mammel and Locke 2019; Cronin et al. 2020). The current study further confirmed RRNC effectiveness by evaluating 285 Manheim Twp homes with pre- and post-RRNC mitigation test results and assessed the radon attributable lung cancer risk reduction associated with RRNC Ordinances.

Our risk estimation relies on applying the dose-response relationships reported by the pooled analysis of European residential case-control studies (Darby et al. 2006). As detailed in the Methods section, this analysis involves using the published linear, non-threshold excess relative risk, relative risk, and absolute risk relationships. We applied these specific risk relationships directly to the average radon concentration reduction measured in the Manheim Twp new dwellings. This approach is a direct application of the model's derived linear factor within the model's established framework and range of residential exposures.

Although the BEIR VI Model was more often used in U.S. radon risk assessments, the Darby model is more appropriate for our analysis because it is directly based on pooled residential data from 13 case–control studies, making it more applicable to typical indoor environment. Also, Darby model provides a robust framework for considering the interaction between radon exposure and smoking status, offering distinct risk estimates for nonsmokers, ex-smokers, and current smokers, which was necessary for incorporating local smoking prevalence data that was available from the PA Department of Health. Nonetheless, the choice of any model involves assumptions. The application of this primarily European-derived model to a US population, while supported by similar findings in North American residential studies, could represent a source of uncertainty in our quantitative risk estimates (Gaskin et al. 2018).

According to the Darby model, lung cancer risks for current smokers had great differences when compared to lifelong nonsmokers and ex-smokers. Smoking is the major cause of lung cancer and the biological mechanisms leading to the development of lung cancer are sub-multiplicatively increased when humans are exposed to radon and are active smokers (Méndez et al. 2011). It should be noted that lung cancer risk estimations in the current study were consistent with the original Darby study. For example, the relative lung cancer risk for current smokers without radon exposure is 25.8, which was very close to our estimation for current smokers in post-mitigation Manheim Twp homes, which was 27.47 (Darby et al. 2005).

As shown in Figure 4B and D, our datasets failed the normality test because of the artificially imposed reporting limit of 0.5 pCi/L. The fact that our data followed a truncated normal distribution indicated that radon test records that were not impacted by the reporting limit are consistent with the findings of previous studies (NRC 1999; Darby et al. 2006). By comparing the average radon levels before and after Tobit regression adjustment, we concluded that Tobit regression was an appropriate methodology to address the left-censoring of the datasets. From Table 1, we can see that censored records rarely appeared in Pre-mitigation records because almost all records with low radon levels were post-mitigation records.

These findings may be limited by the paucity of age stratification information in Manheim Twp. In the Darby model, age-specific dose-response relationships were calculated based on age in some scenarios. This limitation also impacted the estimation of avoided lung cancer death cases because the absolute risks we computed were cumulated to age 75. Additional uncertainty is associated with the smoking prevalence data we obtained from the EDDIE Platform. The Darby model reported divided ex-smokers into different groups based on their quitting time when reporting risks. The EDDIE Platform provided data only on the prevalence of ex-smokers, without details on their quitting times. Consequently, to estimate lung cancer risk among the ex-smoker population, we averaged the risks across all ex-smoker categories. This approach may introduce imprecision in our risk assessment, as it does not account for the varying risks associated with different durations of cessation among ex-smokers.

Our approach to categorize records as pre- or post-mitigation is based on reasonable assumptions about the nature of RRNC and the available data. As mentioned in the Methods section, we selected addresses with two or more tests, then designating the earlier, higher reading as pre-mitigation and the later, 20% or more lower reading as post-mitigation. This rule might misclassify some records when both measurements are in fact post-installation (e.g., seasonal variability) or when a functioning RRNC yields a reduction smaller than 20%. It also excludes new dwellings whose initial radon level was already below 4 pCi/L. In the Mitigation and Raw RRNC datasets, many addresses lacked either a pre- or a post-test, so these addresses were omitted. This decision improved the confidence of our conclusions, but might introduce bias. The Final RRNC dataset contains paired pre- and post-measurements for every address, allowing a true house-by-house comparison; the resulting 16.71 pCi/L drop is therefore our best estimate of RRNC effectiveness.

The successful implementation and effectiveness of such regulations can be influenced by factors beyond the ordinance itself, including the availability of financial resources for homeowners, the local availability of certified professionals and ordinance enforcement. Therefore, while the principles of RRNC are broadly applicable, local context must be carefully considered when evaluating potential effectiveness and implementation strategies elsewhere.

This study provides strong evidence, based on real world data, that the RRNC Ordinances in Manheim Twp have almost eliminated the radon-attributable lung cancer risks in current smokers, ex-smokers, and lifelong nonsmokers. Since Manheim Twp enacted the RRNC Ordinance in 2004, at least 1,733 new single-family homes were built or rebuilt with RRNC features. It can thus be suggested that the RRNC Ordinances at least eliminated radon-attributable lung cancer risks in 10% of Manheim Twp households in the 15-year period, not including the radon mitigation construction that happened in existing houses (U.S. Census Bureau 2022). As EPA reported in 2013, among the 50 states and Washington D.C., only 9 states had statewide RRNC regulations, 12 counties and 19 cities in the other 11 states had local RRNC regulations, while the rest 20 states had neither state nor local RRNC codes (EPA 2013). In total, only 436 counties among the 3144 counties and county-equivalents in the U.S. had requirements for RRNC. Compared with the 1074 counties with average indoor radon levels above EPA's action level, the small portion of counties with RRNC regulations suggests that more assertive legal action is urgently needed to protect against radon's hazards (EPA 2022b). Our study demonstrates that actions like the adoption and effective implementation of RRNC Ordinances in Manheim Twp, represent a promising public health tool with the potential to significantly reduce radon-related lung cancer risk.

Note

1. The study protocol was reviewed by the Johns Hopkins Institutional Review Board and classified as not human subjects research within the meaning of the applicable laws and regulations.

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No potential conflict of interest was reported by the author(s).

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Notes on contributors

Yiguang Zhu is a PhD student at Johns Hopkins Bloomberg School of Public Health with a background in environmental engineering and public health. His research areas include alternatives to animal testing, radon risk, and science-to-policy translation. His doctoral dissertation focuses on the validation and regulatory acceptance of microphysiological systems (organ-on-a-chip). Yiguang earned a B.S. in Environmental System Engineering from Penn State University and an ScM from Johns Hopkins Bloomberg School of Public Health.

Mary A. Fox, PhD, MPH, is an Associate Practice Professor at the Johns Hopkins Bloomberg School of Public Health. She develops and applies cumulative risk assessment methods that consider the combined effects of biological, chemical, environmental, and social factors to inform policies on the environment, worker health and climate adaptation. The concepts of cumulative risk assessment present an opportunity to create interdisciplinary approaches to public and environmental health because of the multiple determinants of health framework that serves as its foundation. Her work addresses the health effects of exposure to chemical mixtures, including whether differential exposure to mixtures contributes to racial and ethnic health disparities. Through her research, she seeks to inform policies that account for multiple chemical and non-chemical exposures.

Bryce Corrigan, PhD, is a Senior Statistician and Lecturer at the SNF Agora Institute at Johns Hopkins University. He specializes in social statistics, focusing on innovative methods for panel data analysis and statistical programming in R, Stata, and Julia. His research examines campaign dynamics and public policy impacts, including cross-sectional analysis of political campaigns and budget policy effects.

Paul A. Locke, an environmental health scientist and attorney, is a Professor at the Johns Hopkins Bloomberg School of Public Health. He holds an MPH from Yale University School of Medicine, a DrPH from the Johns Hopkins University Bloomberg School of Public Health, and a JD degree from Vanderbilt University School of Law. He is admitted to practice law in the state of New York and the District of Columbia, and before the bar of the United States Supreme Court. Dr. Locke's research, practice, and teaching focus on the intersection of law and science, seeking opportunities to replace animals in biomedical research with new, non-animal (in vitro) methods. At Hopkins, he leads an interdisciplinary team of attorneys and scientists who advocate for policy change, primarily in US federal laws and regulations around toxicity testing of environmental chemicals and improvements in the drug development process. In addition, he is an international expert in radiation protection policy and risk communication.

ORCID

Yiguang Zhu http://orcid.org/0000-0002-7954-9105

Data availability statement

Raw data were generated at Pennsylvania Department of Environmental Protection. Derived data supporting the findings of this study are available from the corresponding author YZ on request.

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